

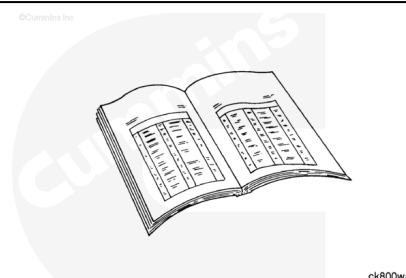
Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

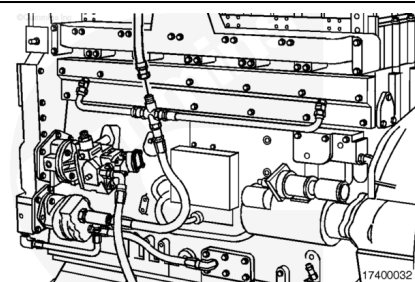
Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on a fuel system.



- Remove the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>).
- Remove the rocker levers. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>).
- Remove the push rods and crossheads. Refer to Procedure 004-014 (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>).
- Remove the fuel supply lines. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html>).
- Remove the crankcase vent tube. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/20/20-003-001-tr.html>).
- Remove the wiring harness. Refer to Procedure 019-043 in the Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666113.html>), or the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html>).

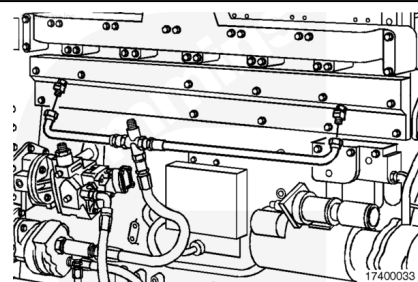
Rail Applications

Disconnect the turbocharger drain hose from the oil scavenge cross connection.



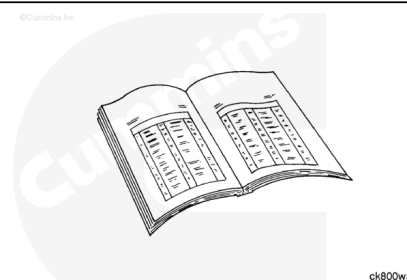
Remove the two oil scavenging tubes from the fittings on the cam follower cover.

Discard the face seal o-rings.



with Electronically Actuated Injector

- Remove the fuel pump. Refer to Procedure 005-016 (</qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html>).
- Remove the fuel manifold. Refer to Procedure 006-022 (</qs3/pubsys2/xml/en/procedures/20/20-006-022-tr.html>).
- Remove the wiring harness. Refer to Procedure 019-043 in the Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666113.html>), or the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html>).
- Remove the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>).
- Remove the rocker lever assembly. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>).

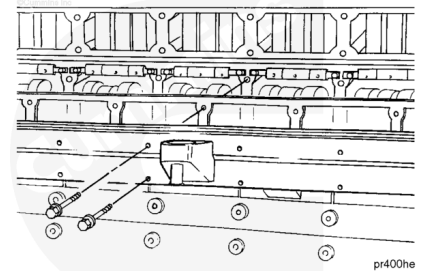


- Remove the push rods or tubes. Refer to Procedure 004-014 (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>).

Remove with Mechanically Actuated Injector

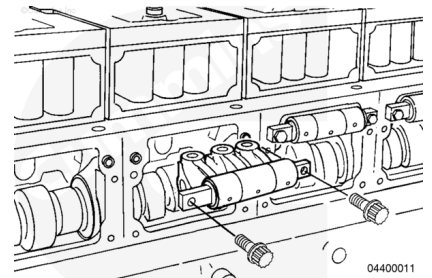
Remove the cam follower cover.

Remove and discard the gasket.



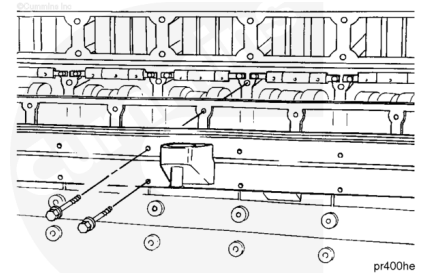
Remove the cam follower assembly capscrews.

Remove the cam follower assembly by pulling straight out until the shaft is off the ring dowels.

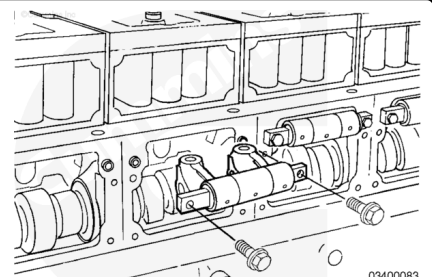


with Electronically Actuated Injector

Remove the cam follower cover. Remove and discard the gasket.



Remove the two capscrews and washers securing the cam follower assembly. Remove the cam follower assembly by pulling straight out until the shaft is off the ring dowels.

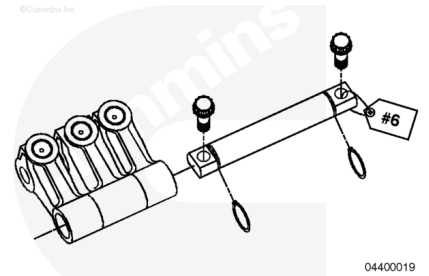


Disassemble

Remove the 12 point capscrews.

Remove the retaining ring. Pull the shaft out.

Mark and tag the parts for future installation identification.

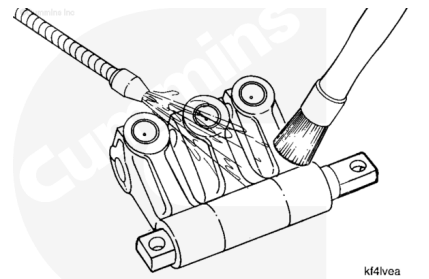


Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

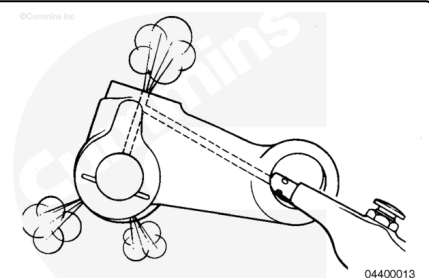
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the cam follower assemblies.

Use compressed air to dry the assemblies.

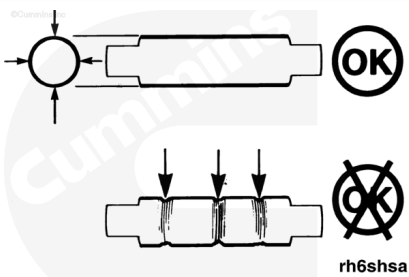
Use solvent to clean the cam followers.

Use compressed air to dry the cam followers. Check the oil drillings to make sure they are **not** plugged or blocked.

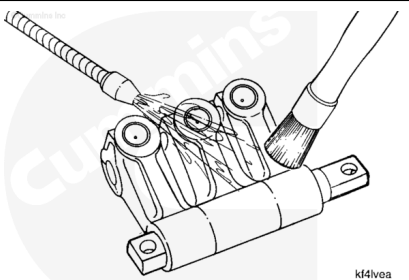


Use solvent to clean the cam follower shaft.

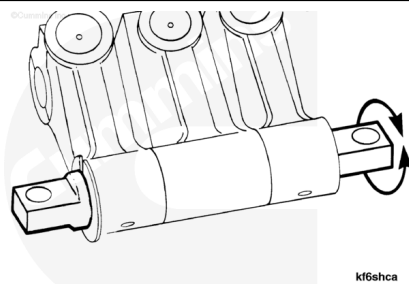
Use compressed air to dry the shaft. Check the nine (9) oil drillings to make sure they are **not** plugged or blocked.



Inspect the cam follower assembly.



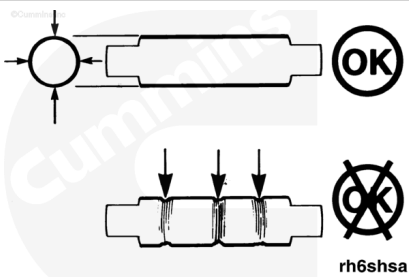
The shaft **must** rotate easily. If resistance is noted, disassemble and check for burrs on the shaft.



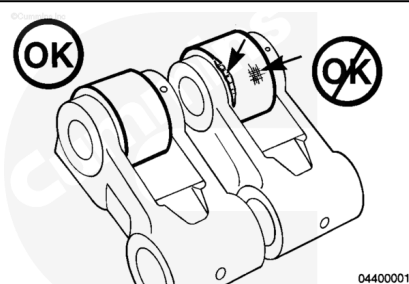
Inspect the cam follower shaft for damage and roughness.

Check the cam follower shaft outside diameter.

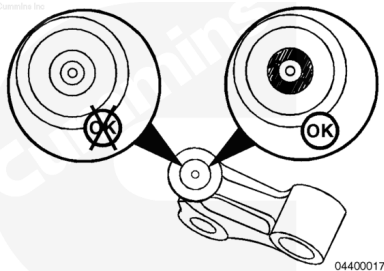
mm		in
28.538	MIN	1.1235
28.575	MAX	1.125



Inspect the cam follower roller for reuse. The cam follower lever assembly **must** be replaced if damaged.



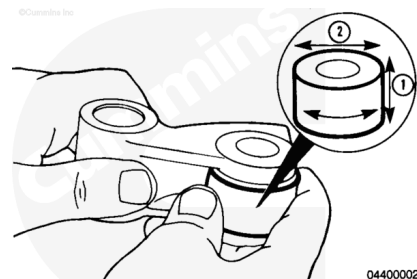
Inspect the push rod socket for damage. If the socket is damaged, the cam follower assembly **must** be replaced.



04400017

The roller **must** rotate easily. Check the clearance of the roller.

	mm		in
1	0.230	MIN	0.0090
	0.610	MAX	0.0240
2	0.076	MIN	0.0030
	0.114	MAX	0.0045

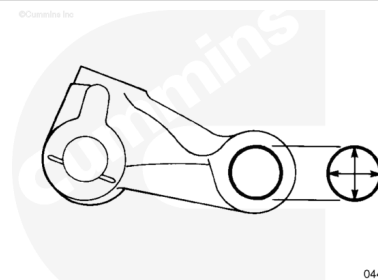


04400002

If the clearances do **not** meet specifications, the cam follower lever assembly **must** be replaced.

Measure the cam follower lever bore inside diameter.

mm		in
28.611	MIN	1.1264
28.661	MAX	1.1284

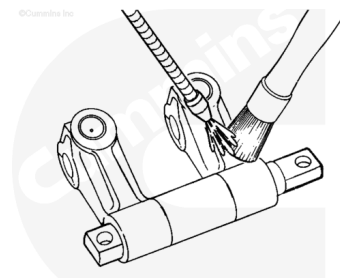


04400018

If the bore inside diameter does **not** meet specifications, the cam follower lever assembly **must** be replaced.

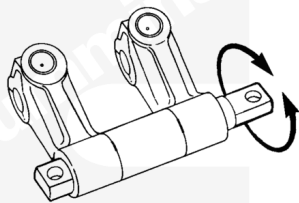
with Electronically Actuated Injector

Clean the cam follower assembly for reuse.



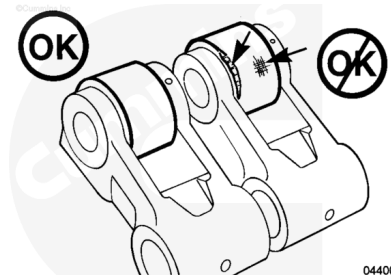
04400030

The shaft **must** rotate freely. If resistance is noted, disassemble and check for burrs on the shaft.



04400031

Inspect the cam follower for reuse. The cam follower lever assembly **must** be replaced if damaged.

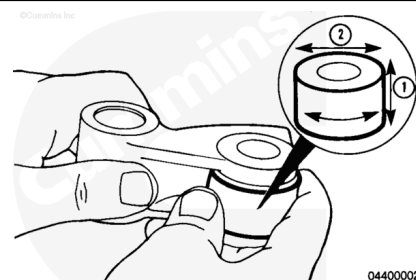


04400001

The roller **must** rotate freely.

Check the clearance of the roller.

	mm		in
1	0.230	MIN	0.0090
	0.610	MAX	0.0240
2	0.076	MIN	0.0030
	0.114	MAX	0.0045



04400002

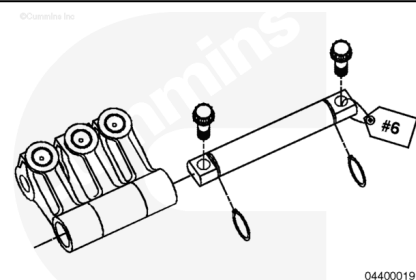
If the clearances do **not** meet specifications, the cam follower lever assembly **must** be replaced.

Assemble

Use clean engine oil to lubricate the shaft and bores of the levers.

Slide the levers on the shaft.

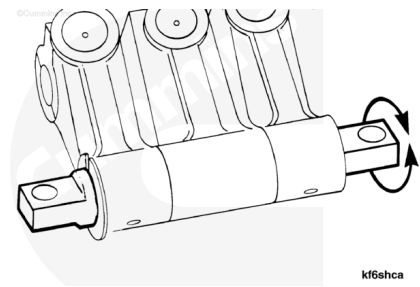
Install the parts illustrated.



04400019

The shaft **must** rotate easily after assembly.

If resistance is apparent, disassemble and check for burrs on the shaft.



Install

with Mechanically Actuated Injector

⚠ CAUTION ⚠

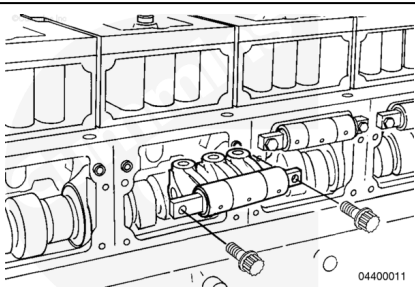
The cam follower mounting capscrews are special 12-point flange heads. Shaft failure will result if standard capscrews are used.

Note : Service replacement cam followers are coated with a heavy preservative to prevent rust. This preservative **must** be removed completely with solvent before the parts are installed on the engine.

Use clean engine oil to lubricate the camshaft and camshaft followers.

Install the cam follower assembly. The shaft **must** fit on both ring dowels.

Tighten the capscrews.

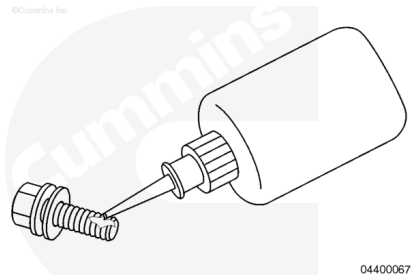


Torque Value:	Step 1	55 n.m [40 ft-lb]
	Step 2	110 n.m [80 ft-lb]
	Step 3	165 n.m [120 ft-lb]

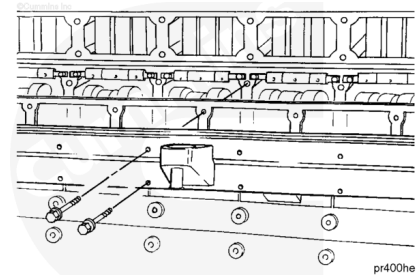
Add a hardened washer to each cam follower cover capscrew.

Apply three drops of Loctite® 246 or equivalent to the capscrews, starting two to three threads from the tip of the capscrew.

Install the gasket, cover, and capscrews.

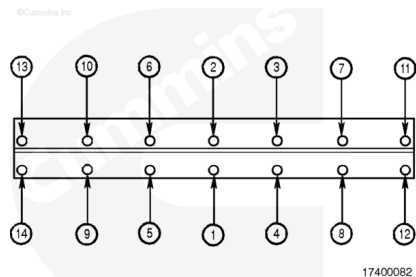


Note : The tabs on the cover gasket must be removed before installation.



Tighten the 14 capscrews in the sequence shown.

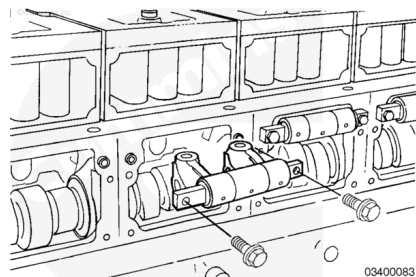
Torque Value: 48 n•m [35 ft-lb]



with Electronically Actuated Injector

⚠ CAUTION ⚠

The cam follower capscrew and washer is special to allow oil to flow to the cam follower shaft. Use of another capscrew will lead to engine damage.



⚠ CAUTION ⚠

The washer is hardened and prevents the capscrew from bottoming out in the hole. Omitting the washer will lead to engine damage.

Note : Service replacement cam followers are coated with a heavy preservative to prevent rust. This preservative must be removed completely with solvent before the parts are installed on the engine. Use clean engine oil to lubricate the cam follower assembly and camshaft.

Install the cam follower assembly. The shaft **must** fit on both ring dowels.

Use clean engine oil to lubricate the capscrew threads.

Install the capscrews and special washers.

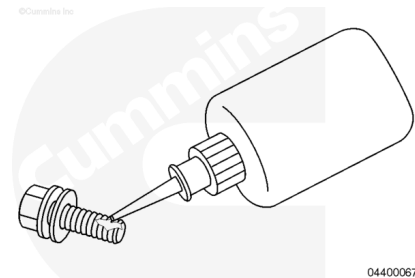
Tighten capscrews.

Torque Value:	Step 1	55 n.m [40 ft-lb]
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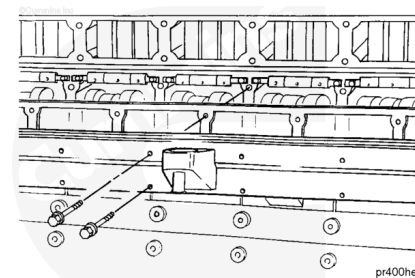
	Step 2	110 n.m [80 ft-lb]
	Step 3	165 n.m [120 ft-lb]

Apply three drops of threadlocker to the cam follower cover capscrews, starting two to three threads from the tip of the capscrew.

Install the cam follower cover gasket, cover, and capscrews.

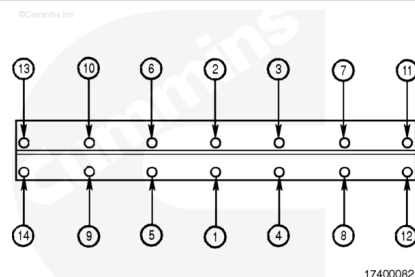


The tabs on the cam follower cover gasket **must** be removed before installation.



Tighten the 14 capscrews in the sequence shown in the illustration.

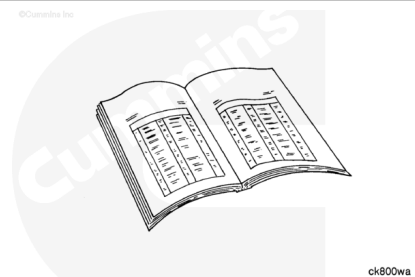
Torque Value: 48 n•m [35 ft-lb]



Finishing Steps

with Mechanically Actuated Injector

- Install the wiring harness. Refer to Procedure 019-043 in the Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666113.html>).
- Install the crankcase vent tube. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/20/20-003-001-tr.html>).

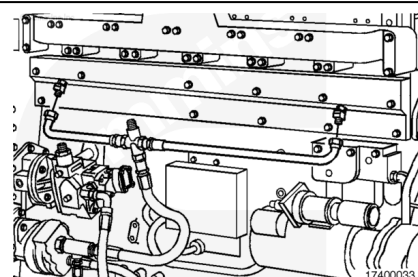


- Install the fuel supply lines. Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html).
- Install the push rods and crossheads. Refer to Procedure 004-014 (/qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html).
- Install the rocker levers. Refer to Procedure 003-009 (/qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html).
- Install the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html).
- Operate the engine and check for leaks.

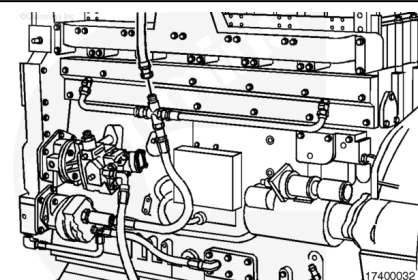
Rail Applications

Install new face seal o-rings.

Install the two oil scavenging tubes on the fittings on the cam follower cover.

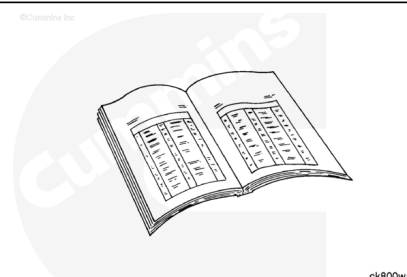


Connect the turbocharger drain hose to the oil scavenge cross.



with Electronically Actuated Injector

- Install push rods or tubes. Refer to Procedure 004-014 (/qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html).
- Install the rocker lever assembly. Refer to Procedure 003-009 (/qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html).
- Install the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html).



- Install the wiring harness. Refer to Procedure 019-043 in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System Series Engine, Bulletin 4021493 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html>).
- Install the fuel manifold. Refer to Procedure 006-022 (</qs3/pubsys2/xml/en/procedures/20/20-006-022-tr.html>).
- Install the fuel pump. Refer to Procedure 005-016 (</qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html>).
- Operate the engine and check for leaks.